

# Contents

|          |                                                                                   |          |
|----------|-----------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Scientific Notes: Energy Pattern Classification of South Atlantic Cyclones</b> | <b>1</b> |
| 1.1      | 1. Introduction                                                                   | 1        |
| 1.2      | 2. Theoretical Background: Lorenz Energy Cycle                                    | 1        |
| 1.3      | 3. Dataset and Filtering                                                          | 3        |
| 1.4      | 4. Methodology                                                                    | 3        |
| 1.5      | 5. Results: Energy Pattern Definitions                                            | 5        |
| 1.6      | 6. Lorenz Phase Space (LPS) Visualization                                         | 6        |
| 1.7      | 7. Scientific Validation                                                          | 7        |
| 1.8      | 8. Computational Implementation                                                   | 7        |
| 1.9      | 9. Conclusions                                                                    | 8        |
| 1.10     | 10. References                                                                    | 8        |

## 1 Scientific Notes: Energy Pattern Classification of South Atlantic Cyclones

**Objective:** Identify and characterize distinct energetic patterns in South Atlantic cyclones through objective clustering analysis.

**Author:** Danilo Couto de Souza

**Date:** February 2026

---

### 1.1 1. Introduction

Extratropical cyclones exhibit diverse energetic behaviors throughout their lifecycle. This analysis identifies distinct **Energy Patterns (EPs)** based on the Lorenz Energy Cycle (LEC) framework, which describes the conversion, generation, and dissipation of energy in atmospheric systems.

**Research Question:** Can South Atlantic cyclones be objectively classified into distinct energy patterns based on their energetic signatures?

**Approach:** Unsupervised clustering (K-Means) applied to Principal Component Analysis (PCA)-reduced energy space.

---

### 1.2 2. Theoretical Background: Lorenz Energy Cycle

The Lorenz Energy Cycle (Lorenz, 1955) partitions atmospheric energy into four reservoirs:

- **Az:** Zonal Available Potential Energy
- **Kz:** Zonal Kinetic Energy
- **Ae:** Eddy Available Potential Energy
- **Ke:** Eddy Kinetic Energy

#### 1.2.1 2.1 Energy Terms Analyzed

This study focuses on **7 energy terms** characterizing cyclone energetics:

| Term       | Name                   | Physical Meaning                                     | Units            |
|------------|------------------------|------------------------------------------------------|------------------|
| <b>Ca</b>  | Baroclinic Conversion  | APE $\rightarrow$ eddy APE via temperature gradients | W m <sup>2</sup> |
| <b>Ck</b>  | Barotropic Conversion  | KE $\rightarrow$ eddy KE via horizontal shear        | W m <sup>2</sup> |
| <b>Ge</b>  | Eddy APE Generation    | Diabatic heating creating eddy APE                   | W m <sup>2</sup> |
| <b>BAe</b> | Eddy APE Boundary Flux | Import/export of eddy APE                            | W m <sup>2</sup> |
| <b>BKe</b> | Eddy KE Boundary Flux  | Import/export of eddy KE                             | W m <sup>2</sup> |
| <b>Ae</b>  | Eddy APE Reservoir     | Available potential energy of eddies                 | J m <sup>2</sup> |
| <b>Ke</b>  | Eddy KE Reservoir      | Kinetic energy of eddies                             | J m <sup>2</sup> |

### 1.2.2 2.2 Key Conversion Terms

#### Baroclinic Conversion (Ca):

$$C_a = \int_{p_b}^{p_t} \frac{R}{p\sigma} \left[ (\omega)_\lambda (T)_\lambda \frac{\partial \langle T \rangle_\lambda}{\partial \phi} + \frac{1}{\sigma} \left[ (\omega)_\lambda (T)_\lambda \frac{\partial \langle \theta \rangle_\lambda}{\partial p} \right]_{\lambda\phi} \right] dp$$

where: -  $\omega$  = vertical velocity (Pa s<sup>-1</sup>) -  $T$  = temperature (K) -  $\theta$  = potential temperature (K) -  $\sigma = -\frac{RT}{p} \frac{\partial \ln \theta}{\partial p}$  = static stability parameter -  $(\cdot)_\lambda$  denotes deviation from zonal mean -  $\langle \cdot \rangle_\lambda$  denotes zonal mean -  $p_t, p_b$  = top and bottom pressure levels

**Physical Interpretation:** - **Ca > 0:** Conversion from zonal APE to eddy APE (baroclinic energy extraction from mean temperature gradients) - **Ca < 0:** Conversion from eddy APE to zonal APE (baroclinic damping)

#### Barotropic Conversion (Ck):

$$C_k = \int_{p_b}^{p_t} \frac{1}{\sigma} \left[ \left[ \frac{\cos \phi}{a} (u)_\lambda (v)_\lambda \frac{\partial}{\partial \phi} \left( \frac{\langle u \rangle_\lambda}{\cos \phi} \right) \right]_{\lambda\phi} + \left[ \frac{(u)_\lambda^2 \partial [u]_\lambda}{a \partial \phi} \right]_{\lambda\phi} + \left[ \frac{\tan \phi}{a} (u)_\lambda^2 [v]_\lambda \right]_{\lambda\phi} + \left[ (\omega)_\lambda (u)_\lambda \frac{\partial [u]_\lambda}{\partial p} \right]_{\lambda\phi} + \left[ (\omega)_\lambda (v)_\lambda \frac{\partial [v]_\lambda}{\partial p} \right]_{\lambda\phi} \right] dp$$

where: -  $u, v$  = zonal and meridional wind components (m s<sup>-1</sup>) -  $a$  = Earth's radius -  $\phi$  = latitude -  $[ \cdot ]_\lambda$  denotes deviation from area mean within LEC domain -  $(\cdot)_\lambda$  denotes deviation from zonal mean

**Physical Interpretation:** -  $C_k < 0$ : Conversion from zonal KE to eddy KE (barotropic energy extraction from mean flow shear) -  $C_k > 0$ : Conversion from eddy KE to zonal KE (barotropic damping)

**Note on sign convention:** - More **negative**  $C_k \rightarrow$  stronger barotropic extraction (eddy gains energy from mean flow) - More **positive**  $C_a \rightarrow$  stronger baroclinic extraction (eddy gains energy from temperature gradients)

---

## 1.3 3. Dataset and Filtering

### 1.3.1 3.1 Original Dataset

**Source:** South Atlantic cyclone tracking algorithm (Reboita et al., 2010; adapted from Hodges, 1994, 1995) - **Total tracks:** 6,789 cyclones - **Temporal coverage:** 1979–2020 (42 years) - **Spatial domain:** 10°S–60°S, 80°W–20°E - **Temporal resolution:** 6-hourly - **Energy calculations:** LEC terms computed using ERA5 reanalysis (Gramscianinov et al., 2020)

### 1.3.2 3.2 Lifecycle Phase Classification

Each cyclone is classified into **4 sequential phases** based on central pressure evolution:

1. **Incipient:** From genesis to onset of deepening
2. **Intensification:** Period of central pressure decrease (deepening)
3. **Mature:** Period of minimum central pressure (near-stationary pressure)
4. **Decay:** Period of central pressure increase (filling)

**Reference:** Sinclair (1994); Reboita et al. (2010)

### 1.3.3 3.3 Quality Control Filtering

To ensure temporal consistency and comparability, only cyclones with **complete lifecycle trajectories** are retained:

**Filtering Criteria:** - Must contain all 4 sequential phases: Incipient  $\rightarrow$  Intensification  $\rightarrow$  Mature  $\rightarrow$  Decay - Each phase must have 1 timestep - Complete energy data for all 7 terms in each phase

**Result:** - **Filtered dataset:** 3,820 cyclones (56.3% of original) - **Total phase records:** 15,280 (3,820 cyclones  $\times$  4 phases) - **Rationale:** Incomplete lifecycles introduce temporal bias; filtering ensures all cyclones are comparable across phases

---

## 1.4 4. Methodology

### 1.4.1 4.1 Step 1: Dimensionality Reduction (PCA)

**Objective:** Reduce 7-dimensional energy space to principal components capturing maximum variance.

**Procedure:**

1. **Normalization:** Standardize each energy term ( $=0$ ,  $=1$ ) using `StandardScaler`
  - Ensures equal weighting regardless of original magnitude
2. **Phase-Separated PCA:** Independent PCA for each lifecycle phase

- **Rationale:** Energy dynamics differ fundamentally across phases; separate PCAs capture phase-specific variance structures
3. **Component Retention:** Keep components explaining 97% cumulative variance
- Typically 6 components per phase
4. **Outputs:**
- PC scores for each cyclone-phase observation
  - Component loadings (energy term contributions to each PC)
  - Explained variance ratios
  - PCA transformation models (saved for reproducibility)

**Result:** Each cyclone-phase observation transformed from 7-dimensional energy space to ~6-dimensional PC space.

#### 1.4.2 4.2 Step 2: Optimal Cluster Number Determination

**Objective:** Objectively determine the number of distinct energy patterns.

**Critical Choice:** All phases **combined** into single dataset (N=15,280) for k-selection - Ensures same number of patterns across all phases (physical consistency) - Larger sample size improves cluster validity statistics

##### Cluster Validity Indices (CVIs):

Five complementary indices computed for  $k = 2$  to 10:

| Index                   | Criterion                                              | Optimal  |
|-------------------------|--------------------------------------------------------|----------|
| Silhouette Coefficient  | Cluster cohesion vs separation                         | Maximize |
| Davies-Bouldin Index    | Within-cluster scatter vs between-cluster separation   | Minimize |
| Calinski-Harabasz Index | Between-cluster variance / within-cluster variance     | Maximize |
| Score Function (SF)     | Intra-cluster compactness and inter-cluster separation | Maximize |
| Gap Statistic           | Compare to null reference distribution                 | Maximize |

**Integration Method:** 1. Normalize all indices to  $[0, 1]$  range 2. Invert indices where “lower is better” (Davies-Bouldin) 3. Compute mean normalized score across all 5 indices 4. Select  $k$  maximizing mean score

**Result:**  $k = 3$  clusters identified as optimal

**Physical Interpretation:** Three distinct energy patterns exist in South Atlantic cyclones.

#### 1.4.3 4.3 Step 3: K-Means Clustering

**Objective:** Assign each cyclone-phase observation to one of three energy patterns.

##### Procedure:

1. **Phase-Separated Clustering:** K-Means ( $k=3$ ) applied independently to each phase’s PC space
  - Respects phase-specific variance structures from PCA
  - Ensures consistency ( $k=3$  for all phases)

2. **Multiple Initializations:** `n_init=100` ensures convergence to global optimum
3. **Centroid Reconstruction:** Transform PC-space centroids back to original energy space
  - Enables physical interpretation of cluster characteristics
  - Uses inverse PCA transformation

**Outputs:** - Cluster assignments for each cyclone-phase observation - Cluster centroids in both PC space and original energy space - Cluster statistics (size, within-cluster variance)

#### 1.4.4 4.4 Step 4: Energy Pattern Characterization

**Objective:** Define Energy Patterns based on cluster centroids and physical interpretation.

**Energy Pattern Assignment:**

Patterns labeled based on **mean Ck** (barotropic conversion) across all phases, sorted from most negative to least negative:

1. **EP1 (Cluster 0):** Lowest (most negative) mean Ck → Strongest barotropic extraction
  2. **EP2 (Cluster 2):** Intermediate mean Ck → Mixed barotropic/baroclinic
  3. **EP3 (Cluster 1):** Highest (least negative) mean Ck → Weakest energetics
- 

### 1.5 5. Results: Energy Pattern Definitions

#### 1.5.1 5.1 EP1: Strong Barotropic and Baroclinic Conversions

**Cluster ID:** 0

**Frequency:** 444 cyclones (11.6% of dataset)

**Mean Ck:** -16.48 W m<sup>2</sup>

**Characteristics:** - **Dominated by strong barotropic and baroclinic conversions** (large negative Ck, high positive Ca) - Pronounced energy extraction from both mean flow and temperature gradients - **Energy exporters:** Negative boundary fluxes indicate energy export to downstream regions - Represents the most energetically active cyclones

**Physical Interpretation:** EP1 cyclones efficiently extract energy from both the mean flow (barotropic) and temperature gradients (baroclinic). Despite intense internal conversions, they tend to **export energy** to surrounding regions, potentially contributing to downstream cyclogenesis and large-scale circulation patterns.

#### 1.5.2 5.2 EP2: Intermediate Barotropic and Baroclinic Conversions

**Cluster ID:** 2

**Frequency:** 979 cyclones (25.6% of dataset)

**Mean Ck:** -3.49 W m<sup>2</sup>

**Characteristics:** - **Moderate and balanced energy conversions** (intermediate Ck and Ca) - Both barotropic and baroclinic processes contribute meaningfully - **Energy importers:** Positive boundary fluxes indicate energy import from surroundings - Represents moderately intense cyclones with external energy coupling

**Physical Interpretation:** EP2 cyclones exhibit balanced baroclinic and barotropic mechanisms with moderate intensity. Critically, they tend to **import energy** from the large-scale environment, suggesting coupling with jet stream dynamics and remote forcing mechanisms.

## Conversion LPS

Figure 1: Conversion LPS

## Imports LPS

Figure 2: Imports LPS

### 1.5.3 5.3 EP3: Weak Energetics

**Cluster ID:** 1

**Frequency:** 2,397 cyclones (62.7% of dataset)

**Mean Ck:**  $-1.71 \text{ W m}^{-2}$

**Characteristics:** - Overall weak energetic conversions (small mean Ck) - Minimal conversion from APE to KE - Lower average intensity compared to EP1/EP2 - Most frequent pattern

**Physical Interpretation:** EP3 represents the “typical” South Atlantic cyclone with relatively weak barotropic and baroclinic activity, consistent with the majority of systems being transient, non-explosive cyclones.

---

## 1.6 6. Lorenz Phase Space (LPS) Visualization

### 1.6.1 6.1 Conversion Phase Space (Ck vs Ca)

The **Conversion LPS** reveals the relationship between baroclinic (Ca) and barotropic (Ck) energy conversions.

**Axes:** - **X-axis (Ck):** Barotropic conversion ( $\text{W m}^{-2}$ ) - More negative  $\rightarrow$  stronger extraction from mean flow - **Y-axis (Ca):** Baroclinic conversion ( $\text{W m}^{-2}$ ) - More positive  $\rightarrow$  stronger extraction from temperature gradients - **Color (Ge):** Eddy APE generation rate ( $\text{W m}^{-2}$ )

**Quadrant Interpretation:** - **Quadrant II (Ck<0, Ca>0):** Classical baroclinic cyclone (both conversions active) - **Quadrant I (Ck>0, Ca>0):** Baroclinic but barotropically damping - **Quadrant III (Ck<0, Ca<0):** Barotropic but baroclinically damping - **Quadrant IV (Ck>0, Ca<0):** Both conversions damping (energy loss)

**Results:**

**Key Findings:** - **EP1** clusters in Quadrant II with large  $|Ck|$  and large Ca  $\rightarrow$  intense baroclinic cyclones - **EP2** shows moderate values in both Ck and Ca  $\rightarrow$  balanced energetics - **EP3** clusters near origin  $\rightarrow$  weak conversions

### 1.6.2 6.2 Boundary Flux Phase Space (BAe vs BKe)

The **Imports LPS** reveals energy import/export through domain boundaries.

**Axes:** - **X-axis (BAe):** Eddy APE boundary flux ( $\text{W m}^{-2}$ ) - Positive  $\rightarrow$  APE import; Negative  $\rightarrow$  APE export - **Y-axis (BKe):** Eddy KE boundary flux ( $\text{W m}^{-2}$ ) - Positive  $\rightarrow$  KE import; Negative  $\rightarrow$  KE export - **Color (Ge):** Eddy APE generation rate ( $\text{W m}^{-2}$ )

**Physical Interpretation:** - Boundary fluxes indicate whether cyclones are energetically isolated or coupled to larger-scale circulation - Positive fluxes  $\rightarrow$  cyclone imports energy from surroundings - Negative fluxes  $\rightarrow$  cyclone exports energy (downstream development)

**Results:**

**Key Findings:** - **EP1** shows **negative boundary fluxes** (Quadrant III/IV) → exports energy to downstream regions - **EP2** shows **positive boundary fluxes** (Quadrant I) → imports energy from jet stream and large-scale environment - **EP3** has weak boundary fluxes near origin → energetically more isolated, minimal external coupling

---

## 1.7 7. Scientific Validation

### 1.7.1 7.1 Cluster Stability

**Within-Cluster Variance:** - EP1: Low variance → cohesive cluster - EP2: Moderate variance → transitional pattern - EP3: Higher variance → diverse “background” pattern

**Cross-Phase Consistency:** EP assignment remains stable across lifecycle phases for most cyclones, indicating robust pattern identification.

### 1.7.2 7.2 Climatological Relevance

#### **Spatial Distribution:**

Based on composite analysis of cyclone genesis locations and tracks (see `scripts/main/06_figure_genesis_de`

- **EP1:** Concentrated in two primary regions:
  1. **Brazil-Malvinas Confluence (BMC):** Most frequent region, associated with strong SST gradients and ocean fronts
  2. **Southeast Brazil coast:** Secondary maximum, potentially including subtropical transition cases
- **EP2:** More widespread distribution across South Atlantic, present in both subtropical and extratropical regions
- **EP3:** Ubiquitous distribution with highest frequency in subtropical latitudes

#### **Seasonal Variation:**

Based on intensity and seasonality analysis (see `scripts/main/05_figure_intensity_seasonality_trends.p`

- **EP1:** Peak in austral winter (JJA) when baroclinicity is strongest
- **EP2:** Year-round occurrence with slight seasonal modulation
- **EP3:** Year-round occurrence with slight summer (DJF) maximum

**Note:** Detailed spatial and temporal distribution figures are available in `figures/main/`.

---

## 1.8 8. Computational Implementation

### 1.8.1 8.1 Software and Packages

**Python 3.13** with scientific stack: - `scikit-learn`: PCA, K-Means, cluster validation - `numpy`, `pandas`: Data manipulation - `matplotlib`, `seaborn`: Visualization - `lorenz-phase-space` v1.3.0: LPS diagram generation

### 1.8.2 8.2 Reproducibility

All analysis scripts include: - Fixed random seeds (`random_state=42`) - Saved PCA transformation models (`results/cluster_analysis_energy_patterns/pca_models/`) - Saved K-Means

models(results/cluster\_analysis\_energy\_patterns/kmeans\_models/) - Version-controlled codebase

### Pipeline Execution:

```
python scripts/cluster_analysis_energy_patterns/run_all.py
```

### Individual Steps:

```
python scripts/cluster_analysis_energy_patterns/step1_normalize_and_pca.py
python scripts/cluster_analysis_energy_patterns/step2_plot_pca_results.py
python scripts/cluster_analysis_energy_patterns/step3_optimal_k_analysis.py
python scripts/cluster_analysis_energy_patterns/step4_apply_kmeans.py
python scripts/cluster_analysis_energy_patterns/step5_plot_energy_patterns.py
```

---

## 1.9 9. Conclusions

### 1.9.1 9.1 Main Findings

1. **Three distinct energy patterns** objectively identified through multivariate clustering
2. **EP1 (11.6%)**: Intense barotropic and baroclinic cyclones
3. **EP2 (25.6%)**: Moderate energetics with balanced conversions
4. **EP3 (62.7%)**: Weak energetics representing typical cyclone behavior

### 1.9.2 9.2 Physical Significance

The energy pattern classification reveals that: - South Atlantic cyclones span a continuum from **intense** (EP1) to **weak** (EP3) energetic activity - **Barotropic conversion (Ck)** is the primary discriminating factor - Baroclinic processes (Ca) remain important across all patterns but with varying intensity

### 1.9.3 9.3 Implications

**For cyclone dynamics:** - Energy patterns reflect different intensification mechanisms - EP1 cyclones likely associated with explosive development and high societal impact

**For predictability:** - Different energy patterns may have different forecast skill - Pattern-specific model biases possible

**For climate studies:** - Energy pattern frequency may shift under climate change - Provides objective framework for inter-model comparison

---

## 1.10 10. References

1. **Lorenz, E. N. (1955)**. Available potential energy and the maintenance of the general circulation. *Tellus*, 7(2), 157-167. DOI: 10.3402/tellusa.v7i2.8796
2. **Gramscianinov, C. B., Campos, R. M., de Camargo, R., Hodges, K. I., Guedes Soares, C., & da Silva Dias, P. L. (2020)**. Analysis of Atlantic extratropical storm tracks characteristics in 41 years of ERA5 and CFSR/CFSv2 databases. *Ocean Engineering*, 216, 108111. DOI: 10.1016/j.oceaneng.2020.108111



3. **Reboita, M. S., da Rocha, R. P., Ambrizzi, T., & Sugahara, S. (2010).** South Atlantic Ocean cyclogenesis climatology simulated by regional climate model (RegCM3). *Climate Dynamics*, 35, 1331-1347. DOI: 10.1007/s00382-009-0668-7
4. **Hodges, K. I. (1994).** A general method for tracking analysis and its application to meteorological data. *Monthly Weather Review*, 122(11), 2573-2586.
5. **Hodges, K. I. (1995).** Feature tracking on the unit sphere. *Monthly Weather Review*, 123(12), 3458-3465.
6. **Sinclair, M. R. (1994).** An objective cyclone climatology for the Southern Hemisphere. *Monthly Weather Review*, 122(10), 2239-2256.
7. **Tibshirani, R., Walther, G., & Hastie, T. (2001).** Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B*, 63(2), 411-423.
8. **Rousseeuw, P. J. (1987).** Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53-65.

---

**Document Version:** 1.0

**Last Updated:** February 23, 2026